

Transport of Substances Through the Cell Membrane



Figure 4–1 gives the approximate concentrations of important electrolytes and other substances in the *extracellular fluid* and *intracellular fluid*. Note that the extracellular fluid contains a large amount of *sodium* but only a small amount of *potassium*. Exactly the opposite is true of the intracellular fluid. Also, the extracellular fluid contains a large amount of chloride ions, whereas the intracellular fluid con-

tains very little. But the concentrations of phosphates and proteins in the intracellular fluid are considerably greater than those in the extracellular fluid. These differences are extremely important to the life of the cell. The purpose of this chapter is to explain how the differences are brought about by the transport mechanisms of the cell membranes.

The Lipid Barrier of the Cell Membrane, and Cell Membrane Transport Proteins

The structure of the membrane covering the outside of every cell of the body is discussed in Chapter 2 and illustrated in Figures 2–3 and 4–2. This membrane consists almost entirely of a *lipid bilayer*, but it also contains large numbers of protein molecules in the lipid, many of which penetrate all the way through the membrane, as shown in Figure 4–2.

The lipid bilayer is not miscible with either the extracellular fluid or the intracellular fluid. Therefore, it constitutes a barrier against movement of water molecules and water-soluble substances between the extracellular and intracellular fluid compartments. However, as demonstrated in Figure 4–2 by the left-most arrow, a few substances can penetrate this lipid bilayer, diffusing directly through the lipid substance itself; this is true mainly of lipid-soluble substances, as described later.

The protein molecules in the membrane have entirely different properties for transporting substances. Their molecular structures interrupt the continuity of the lipid bilayer, constituting an alternative pathway through the cell membrane. Most of these penetrating proteins, therefore, can function as *transport proteins*. Different proteins function differently. Some have watery spaces all the way through the molecule and allow free movement of water as well as selected ions or molecules; these are called *channel proteins*. Others, called *carrier proteins*, bind with molecules or ions that are to be transported; conformational changes in the protein molecules then move the substances through the interstices of the protein to the other side of the membrane. Both the channel proteins and the carrier proteins are usually highly selective in the types of molecules or ions that are allowed to cross the membrane.

“Diffusion” Versus “Active Transport.” Transport through the cell membrane, either directly through the lipid bilayer or through the proteins, occurs by one of two basic processes: *diffusion* or *active transport*.

Although there are many variations of these basic mechanisms, diffusion means random molecular movement of substances molecule by molecule, either through intermolecular spaces in the membrane or in combination with a carrier

EXTRACELLULAR FLUID		INTRACELLULAR FLUID	
Na ⁺	142 mEq/L	10 mEq/L	
K ⁺	4 mEq/L	140 mEq/L	
Ca ⁺⁺	2.4 mEq/L	0.0001 mEq/L	
Mg ⁺⁺	1.2 mEq/L	58 mEq/L	
Cl ⁻	103 mEq/L	4 mEq/L	
HCO ₃ ⁻	28 mEq/L	10 mEq/L	
Phosphates	4 mEq/L	75 mEq/L	
SO ₄ ⁻	1 mEq/L	2 mEq/L	
Glucose	90 mg/dl	0 to 20 mg/dl	
Amino acids	30 mg/dl	200 mg/dl ?	
Cholesterol	0.5 g/dl	2 to 95 g/dl	
Phospholipids			
Neutral fat			
PO ₂	35 mm Hg	20 mm Hg ?	
PCO ₂	46 mm Hg	50 mm Hg ?	
pH	7.4	7.0	
Proteins	2 g/dl	16 g/dl	
	(5 mEq/L)	(40 mEq/L)	

Figure 4-1

Chemical compositions of extracellular and intracellular fluids.

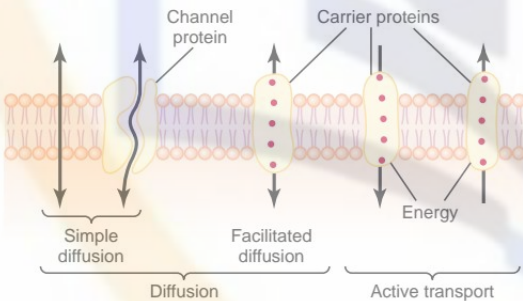


Figure 4-2

Transport pathways through the cell membrane, and the basic mechanisms of transport.

protein. The energy that causes diffusion is the energy of the normal kinetic motion of matter.

By contrast, active transport means movement of ions or other substances across the membrane in combination with a carrier protein in such a way that the carrier protein causes the substance to move against an energy gradient, such as from a low-concentration state to a high-concentration state. This movement requires an additional source of energy besides kinetic energy. Following is a more detailed explanation of



Figure 4-3

Diffusion of a fluid molecule during a thousandth of a second.

the basic physics and physical chemistry of these two processes.

Diffusion

All molecules and ions in the body fluids, including water molecules and dissolved substances, are in constant motion, each particle moving its own separate way. Motion of these particles is what physicists call “heat”—the greater the motion, the higher the temperature—and the motion never ceases under any condition except at absolute zero temperature. When a moving molecule, A, approaches a stationary molecule, B, the electrostatic and other nuclear forces of molecule A repel molecule B, transferring some of the energy of motion of molecule A to molecule B. Consequently, molecule B gains kinetic energy of motion, while molecule A slows down, losing some of its kinetic energy. Thus, as shown in Figure 4-3, a single molecule in a solution bounces among the other molecules first in one direction, then another, then another, and so forth, randomly bouncing thousands of times each second. This continual movement of molecules among one another in liquids or in gases is called *diffusion*.

Ions diffuse in the same manner as whole molecules, and even suspended colloid particles diffuse in a similar manner, except that the colloids diffuse far less rapidly than molecular substances because of their large size.

Diffusion Through the Cell Membrane

Diffusion through the cell membrane is divided into two subtypes called *simple diffusion* and *facilitated diffusion*. Simple diffusion means that kinetic movement of molecules or ions occurs through a membrane opening or through intermolecular spaces without any interaction with carrier proteins in the membrane. The rate of diffusion is determined by the amount of

substance available, the velocity of kinetic motion, and the number and sizes of openings in the membrane through which the molecules or ions can move.

Facilitated diffusion requires interaction of a carrier protein. The carrier protein aids passage of the molecules or ions through the membrane by binding chemically with them and shuttling them through the membrane in this form.

Simple diffusion can occur through the cell membrane by two pathways: (1) through the interstices of the lipid bilayer if the diffusing substance is lipid soluble, and (2) through watery channels that penetrate all the way through some of the large transport proteins, as shown to the left in Figure 4–2.

Diffusion of Lipid-Soluble Substances Through the Lipid Bilayer.

One of the most important factors that determines how rapidly a substance diffuses through the lipid bilayer is the *lipid solubility* of the substance. For instance, the lipid solubilities of oxygen, nitrogen, carbon dioxide, and alcohols are high, so that all these can dissolve directly in the lipid bilayer and diffuse through the cell membrane in the same manner that diffusion of water solutes occurs in a watery solution. For obvious reasons, the rate of diffusion of each of these substances through the membrane is directly proportional to its lipid solubility. Especially large amounts of oxygen can be transported in this way; therefore, oxygen can be delivered to the interior of the cell almost as though the cell membrane did not exist.

Diffusion of Water and Other Lipid-Insoluble Molecules Through Protein Channels. Even though water is highly insoluble in the membrane lipids, it readily passes through channels in protein molecules that penetrate all the way through the membrane. The rapidity with which water molecules can move through most cell membranes is astounding. As an example, the total amount of water that diffuses in each direction through the red cell membrane during each second is about 100 times as great as the volume of the red cell itself.

Other lipid-insoluble molecules can pass through the protein pore channels in the same way as water molecules if they are water soluble and small enough. However, as they become larger, their penetration falls off rapidly. For instance, the diameter of the urea molecule is only 20 per cent greater than that of water, yet its penetration through the cell membrane pores is about 1000 times less than that of water. Even so, given the astonishing rate of water penetration, this amount of urea penetration still allows rapid transport of urea through the membrane within minutes.

Diffusion Through Protein Channels, and “Gating” of These Channels

Computerized three-dimensional reconstructions of protein channels have demonstrated tubular pathways all the way from the extracellular to the intracellular fluid. Therefore, substances can move by simple

diffusion directly along these channels from one side of the membrane to the other. The protein channels are distinguished by two important characteristics: (1) they are often selectively permeable to certain substances, and (2) many of the channels can be opened or closed by *gates*.

Selective Permeability of Protein Channels. Many of the protein channels are highly selective for transport of one or more specific ions or molecules. This results from the characteristics of the channel itself, such as its diameter, its shape, and the nature of the electrical charges and chemical bonds along its inside surfaces. To give an example, one of the most important of the protein channels, the so-called *sodium channel*, is only 0.3 by 0.5 nanometer in diameter, but more important, the inner surfaces of this channel are *strongly negatively charged*, as shown by the negative signs inside the channel proteins in the top panel of Figure 4–4. These strong negative charges can pull small *dehydrated* sodium ions into these channels, actually pulling the sodium ions away from their hydrating water molecules. Once in the channel, the sodium ions diffuse in either direction according to the usual laws of diffusion. Thus, the sodium channel is specifically selective for passage of sodium ions.

Conversely, another set of protein channels is selective for potassium transport, shown in the lower panel of Figure 4–4. These channels are slightly smaller than the sodium channels, only 0.3 by 0.3 nanometer, but *they are not negatively charged*, and their chemical bonds are different. Therefore, no strong attractive force is pulling ions into the channels, and the potassium ions are not pulled away from the water

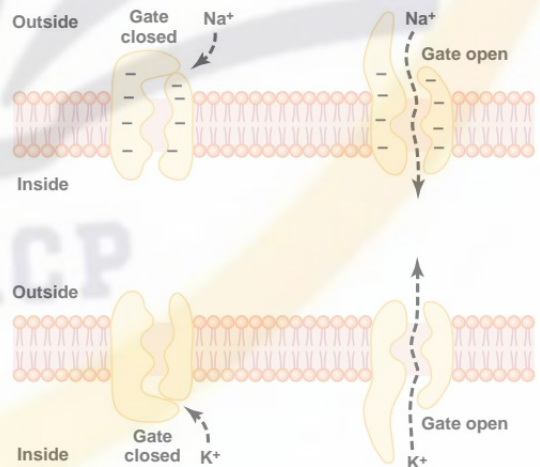


Figure 4–4

Transport of sodium and potassium ions through protein channels. Also shown are conformational changes in the protein molecules to open or close “gates” guarding the channels.

molecules that hydrate them. The hydrated form of the potassium ion is considerably smaller than the hydrated form of sodium because the sodium ion attracts far more water molecules than does potassium. Therefore, the smaller hydrated potassium ions can pass easily through this small channel, whereas the larger hydrated sodium ions are rejected, thus providing selective permeability for a specific ion.

Gating of Protein Channels. Gating of protein channels provides a means of controlling ion permeability of the channels. This is shown in both panels of Figure 4-4 for selective gating of sodium and potassium ions. It is believed that some of the gates are actual gatelike extensions of the transport protein molecule, which can close the opening of the channel or can be lifted away from the opening by a conformational change in the shape of the protein molecule itself.

The opening and closing of gates are controlled in two principal ways:

1. **Voltage gating.** In this instance, the molecular conformation of the gate or of its chemical bonds responds to the electrical potential across the cell membrane. For instance, in the top panel of Figure 4-4, when there is a strong negative charge on the inside of the cell membrane, this presumably could cause the outside sodium gates to remain tightly closed; conversely, when the inside of the membrane loses its negative charge, these gates would open suddenly and allow tremendous quantities of sodium to pass inward through the sodium pores. This is the basic mechanism for eliciting action potentials in nerves that are responsible for nerve signals. In the bottom panel of Figure 4-4, the potassium gates are on the intracellular ends of the potassium channels, and they open when the inside of the cell membrane becomes positively charged. The opening of these gates is partly responsible for terminating the action potential, as is discussed more fully in Chapter 5.
2. **Chemical (ligand) gating.** Some protein channel gates are opened by the binding of a chemical substance (a ligand) with the protein; this causes a conformational or chemical bonding change in the protein molecule that opens or closes the gate. This is called *chemical gating* or *ligand gating*. One of the most important instances of chemical gating is the effect of acetylcholine on the so-called *acetylcholine channel*. Acetylcholine opens the gate of this channel, providing a negatively charged pore about 0.65 nanometer in diameter that allows uncharged molecules or positive ions smaller than this diameter to pass through. This gate is exceedingly important for the transmission of nerve signals from one nerve cell to another (see Chapter 45) and from nerve cells to muscle cells to cause muscle contraction (see Chapter 7).

Open-State Versus Closed-State of Gated Channels.

Figure 4-5A shows an especially interesting characteristic of most voltage-gated channels. This figure shows two recordings of electrical current flowing through a single sodium channel when there was an approximate 25-millivolt potential gradient across the

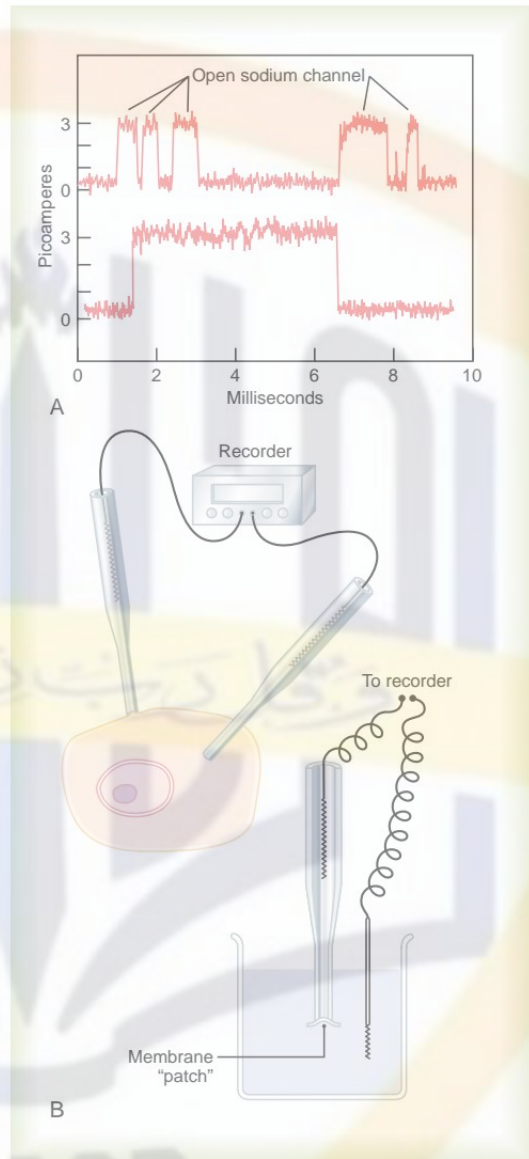


Figure 4-5

A, Record of current flow through a single voltage-gated sodium channel, demonstrating the "all or none" principle for opening and closing of the channel. B, The "patch-clamp" method for recording current flow through a single protein channel. To the left, recording is performed from a "patch" of a living cell membrane. To the right, recording is from a membrane patch that has been torn away from the cell.

membrane. Note that the channel conducts current either "all or none." That is, the gate of the channel snaps open and then snaps closed, each open state lasting for only a fraction of a millisecond up to several milliseconds. This demonstrates the rapidity with

which changes can occur during the opening and closing of the protein molecular gates. At one voltage potential, the channel may remain closed all the time or almost all the time, whereas at another voltage level, it may remain open either all or most of the time. At in-between voltages, as shown in the figure, the gates tend to snap open and closed intermittently, giving an average current flow somewhere between the minimum and the maximum.

Patch-Clamp Method for Recording Ion Current Flow Through Single Channels. One might wonder how it is technically possible to record ion current flow through single protein channels as shown in Figure 4-5A. This has been achieved by using the “patch-clamp” method illustrated in Figure 4-5B. Very simply, a micropipette, having a tip diameter of only 1 or 2 micrometers, is abutted against the outside of a cell membrane. Then suction is applied inside the pipette to pull the membrane against the tip of the pipette. This creates a seal where the edges of the pipette touch the cell membrane. The result is a minute membrane “patch” at the tip of the pipette through which electrical current flow can be recorded.

Alternatively, as shown to the right in Figure 4-5B, the small cell membrane patch at the end of the pipette can be torn away from the cell. The pipette with its sealed patch is then inserted into a free solution. This allows the concentrations of ions both inside the micropipette and in the outside solution to be altered as desired. Also, the voltage between the two sides of the membrane can be set at will—that is, “clamped” to a given voltage.

It has been possible to make such patches small enough so that only a single channel protein is found in the membrane patch being studied. By varying the concentrations of different ions, as well as the voltage across the membrane, one can determine the transport characteristics of the single channel and also its gating properties.

Facilitated Diffusion

Facilitated diffusion is also called *carrier-mediated diffusion* because a substance transported in this manner diffuses through the membrane using a specific carrier protein to help. That is, the carrier *facilitates* diffusion of the substance to the other side.

Facilitated diffusion differs from simple diffusion in the following important way: Although the rate of simple diffusion through an open channel increases proportionately with the concentration of the diffusing substance, in facilitated diffusion the rate of diffusion approaches a maximum, called $V_{\max}$, as the concentration of the diffusing substance increases. This difference between simple diffusion and facilitated diffusion is demonstrated in Figure 4-6. The figure shows that as the concentration of the diffusing substance increases, the rate of simple diffusion continues to increase proportionately, but in the case of facilitated diffusion, the rate of diffusion cannot rise greater than the $V_{\max}$ level.

What is it that limits the rate of facilitated diffusion? A probable answer is the mechanism illustrated in Figure 4-7. This figure shows a carrier protein with a

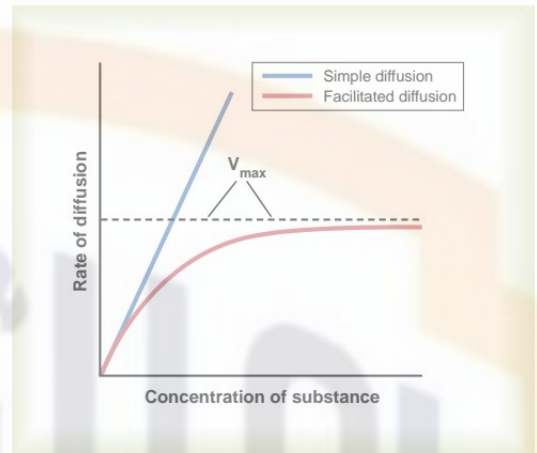


Figure 4-6

Effect of concentration of a substance on rate of diffusion through a membrane by simple diffusion and facilitated diffusion. This shows that facilitated diffusion approaches a maximum rate called the $V_{\max}$.

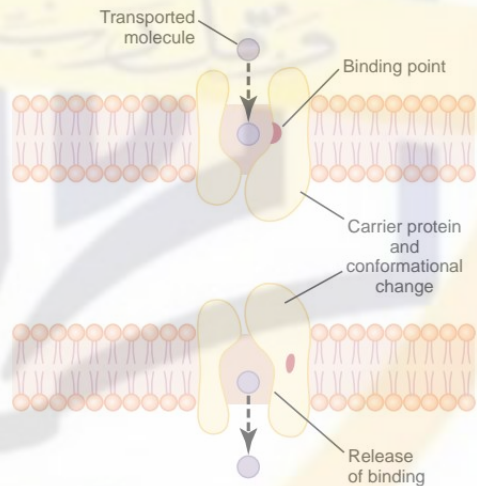


Figure 4-7

Postulated mechanism for facilitated diffusion.

pore large enough to transport a specific molecule partway through. It also shows a binding “receptor” on the inside of the protein carrier. The molecule to be transported enters the pore and becomes bound. Then, in a fraction of a second, a conformational or chemical change occurs in the carrier protein, so that the pore now opens to the opposite side of the membrane. Because the binding force of the receptor is weak, the thermal motion of the attached molecule causes it to break away and to be released on the opposite side of

the membrane. The rate at which molecules can be transported by this mechanism can never be greater than the rate at which the carrier protein molecule can undergo change back and forth between its two states. Note specifically, though, that this mechanism allows the transported molecule to move—that is, to “diffuse”—in either direction through the membrane.

Among the most important substances that cross cell membranes by facilitated diffusion are *glucose* and most of the *amino acids*. In the case of glucose, the carrier molecule has been discovered, and it has a molecular weight of about 45,000; it can also transport several other monosaccharides that have structures similar to that of glucose, including galactose. Also, insulin can increase the rate of facilitated diffusion of glucose as much as 10-fold to 20-fold. This is the principal mechanism by which insulin controls glucose use in the body, as discussed in Chapter 78.

Factors That Affect Net Rate of Diffusion

By now it is evident that many substances can diffuse through the cell membrane. What is usually important is the *net* rate of diffusion of a substance in the desired direction. This net rate is determined by several factors.

Effect of Concentration Difference on Net Diffusion Through a Membrane. Figure 4–8A shows a cell membrane with a substance in high concentration on the outside and low concentration on the inside. The rate at which the substance diffuses *inward* is proportional to the concentration of molecules on the *outside*, because this concentration determines how many molecules strike the outside of the membrane each second. Conversely, the rate at which molecules diffuse *outward* is proportional to their concentration *inside* the membrane. Therefore, the rate of net diffusion into the cell is proportional to the concentration on the outside *minus* the concentration on the inside, or:

$$\text{Net diffusion} \propto (C_o - C_i)$$

in which C_o is concentration outside and C_i is concentration inside.

Effect of Membrane Electrical Potential on Diffusion of Ions—The “Nernst Potential.” If an electrical potential is applied across the membrane, as shown in Figure 4–8B, the electrical charges of the ions cause them to move through the membrane even though no concentration difference exists to cause movement. Thus, in the left panel of Figure 4–8B, the concentration of *negative* ions is the same on both sides of the membrane, but a positive charge has been applied to the right side of the membrane and a negative charge to the left, creating an electrical gradient across the membrane. The positive charge attracts the negative ions, whereas the negative charge repels them. Therefore, net diffusion occurs from left to right. After much time, large quantities of negative ions have moved to the

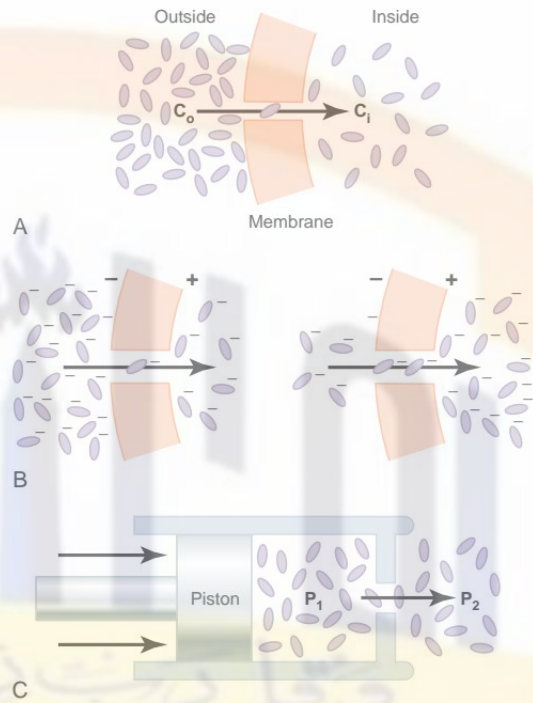


Figure 4–8

Effect of concentration difference (A), electrical potential difference affecting negative ions (B), and pressure difference (C) to cause diffusion of molecules and ions through a cell membrane.

right, creating the condition shown in the right panel of Figure 4–8B, in which a concentration difference of the ions has developed in the direction opposite to the electrical potential difference. The concentration difference now tends to move the ions to the left, while the electrical difference tends to move them to the right. When the concentration difference rises high enough, the two effects balance each other. At normal body temperature (37°C), the electrical difference that will balance a given concentration difference of *univalent* ions—such as sodium (Na^+) ions—can be determined from the following formula, called the *Nernst equation*:

$$\text{EMF (in millivolts)} = \pm 61 \log \frac{C_1}{C_2}$$

in which EMF is the electromotive force (voltage) between side 1 and side 2 of the membrane, C_1 is the concentration on side 1, and C_2 is the concentration on side 2. This equation is extremely important in understanding the transmission of nerve impulses and is discussed in much greater detail in Chapter 5.

Effect of a Pressure Difference Across the Membrane. At times, considerable pressure difference develops

between the two sides of a diffusible membrane. This occurs, for instance, at the blood capillary membrane in all tissues of the body. The pressure is about 20 mm Hg greater inside the capillary than outside.

Pressure actually means the sum of all the forces of the different molecules striking a unit surface area at a given instant. Therefore, when the pressure is higher on one side of a membrane than on the other, this means that the sum of all the forces of the molecules striking the channels on that side of the membrane is greater than on the other side. In most instances, this is caused by greater numbers of molecules striking the membrane per second on one side than on the other side. The result is that increased amounts of energy are available to cause net movement of molecules from the high-pressure side toward the low-pressure side. This effect is demonstrated in Figure 4-8C, which shows a piston developing high pressure on one side of a “pore,” thereby causing more molecules to strike the pore on this side and, therefore, more molecules to “diffuse” to the other side.

Osmosis Across Selectively Permeable Membranes—“Net Diffusion” of Water

By far the most abundant substance that diffuses through the cell membrane is water. Enough water ordinarily diffuses in each direction through the red cell membrane per second to equal about *100 times the volume of the cell itself*. Yet, normally, the amount that diffuses in the two directions is balanced so precisely that zero *net* movement of water occurs. Therefore, the volume of the cell remains constant. However, under certain conditions, a *concentration difference for water* can develop across a membrane, just as concentration differences for other substances can occur. When this happens, net movement of water does occur across the cell membrane, causing the cell either to swell or to shrink, depending on the direction of the water movement. This process of net movement of water caused by a concentration difference of water is called *osmosis*.

To give an example of osmosis, let us assume the conditions shown in Figure 4-9, with pure water on one side of the cell membrane and a solution of sodium chloride on the other side. Water molecules pass through the cell membrane with ease, whereas sodium and chloride ions pass through only with difficulty. Therefore, sodium chloride solution is actually a mixture of permeant water molecules and nonpermeant sodium and chloride ions, and the membrane is said to be *selectively permeable* to water but much less so to sodium and chloride ions. Yet the presence of the sodium and chloride has displaced some of the water molecules on the side of the membrane where these ions are present and, therefore, has reduced the concentration of water molecules to less than that of pure water. As a result, in the example of Figure 4-9, more water molecules strike the channels on the left side,

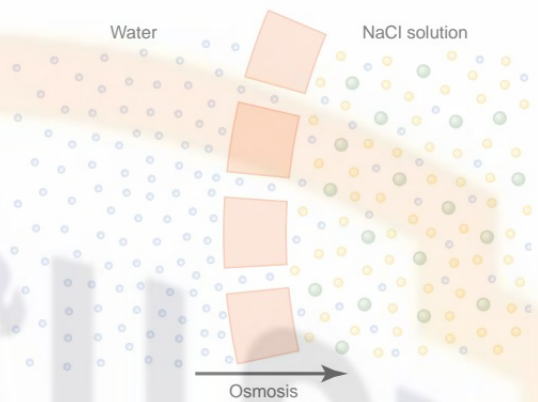


Figure 4-9

Osmosis at a cell membrane when a sodium chloride solution is placed on one side of the membrane and water is placed on the other side.

where there is pure water, than on the right side, where the water concentration has been reduced. Thus, net movement of water occurs from left to right—that is, *osmosis* occurs from the pure water into the sodium chloride solution.

Osmotic Pressure

If in Figure 4-9 pressure were applied to the sodium chloride solution, osmosis of water into this solution would be slowed, stopped, or even reversed. The exact amount of pressure required to stop osmosis is called the *osmotic pressure* of the sodium chloride solution.

The principle of a pressure difference opposing osmosis is demonstrated in Figure 4-10, which shows a selectively permeable membrane separating two columns of fluid, one containing pure water and the other containing a solution of water and any solute that will not penetrate the membrane. Osmosis of water from chamber B into chamber A causes the levels of the fluid columns to become farther and farther apart, until eventually a pressure difference develops between the two sides of the membrane great enough to oppose the osmotic effect. The pressure difference across the membrane at this point is equal to the osmotic pressure of the solution that contains the nondiffusible solute.

Importance of Number of Osmotic Particles (Molar Concentration) in Determining Osmotic Pressure. The osmotic pressure exerted by particles in a solution, whether they are molecules or ions, is determined by the *number* of particles per unit volume of fluid, *not by the mass* of the particles. The reason for this is that each particle in a solution, regardless of its mass, exerts, on average, the same amount of pressure against the membrane. That is, large particles, which have greater mass (*m*) than small particles, move at slower velocities (*v*). The small particles move at higher velocities in such a way

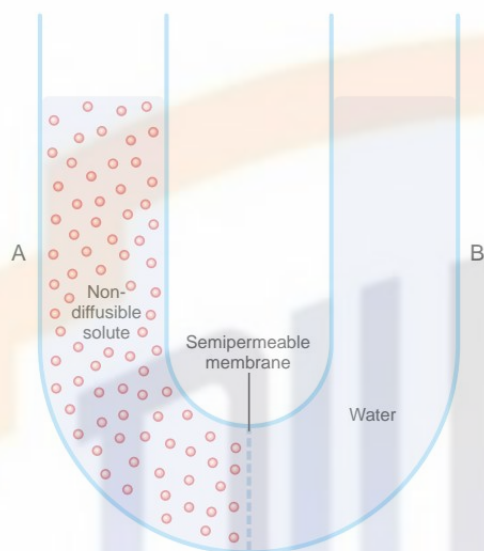


Figure 4-10

Demonstration of osmotic pressure caused by osmosis at a semipermeable membrane.

that their average kinetic energies (k), determined by the equation

$$k = \frac{mv^2}{2}$$

are the same for each small particle as for each large particle. Consequently, the factor that determines the osmotic pressure of a solution is the concentration of the solution in terms of number of particles (which is the same as its *molar concentration* if it is a nondissociated molecule), not in terms of mass of the solute.

“Osmolality”—The Osmole. To express the concentration of a solution in terms of numbers of particles, the unit called the *osmole* is used in place of grams.

One osmole is 1 gram molecular weight of osmotically active solute. Thus, 180 grams of glucose, which is 1 gram molecular weight of glucose, is equal to 1 osmole of glucose because glucose does not dissociate into ions. Conversely, if a solute dissociates into two ions, 1 gram molecular weight of the solute will become 2 osmoles because the number of osmotically active particles is now twice as great as is the case for the nondissociated solute. Therefore, when fully dissociated, 1 gram molecular weight of sodium chloride, 58.5 grams, is equal to 2 osmoles.

Thus, a solution that has 1 osmole of solute dissolved in each kilogram of water is said to have an *osmolality* of 1 osmole per kilogram, and a solution that has 1/1000 osmole dissolved per kilogram has an osmolality of 1 milliosmole per kilogram. The normal

osmolality of the extracellular and intracellular fluids is about 300 milliosmoles per kilogram of water.

Relation of Osmolality to Osmotic Pressure. At normal body temperature, 37°C, a concentration of 1 osmole per liter will cause 19,300 mm Hg osmotic pressure in the solution. Likewise, 1 milliosmole per liter concentration is equivalent to 19.3 mm Hg osmotic pressure. Multiplying this value by the 300 milliosmolar concentration of the body fluids gives a total calculated osmotic pressure of the body fluids of 5790 mm Hg. The measured value for this, however, averages only about 5500 mm Hg. The reason for this difference is that many of the ions in the body fluids, such as sodium and chloride ions, are highly attracted to one another; consequently, they cannot move entirely unrestrained in the fluids and create their full osmotic pressure potential. Therefore, on average, the actual osmotic pressure of the body fluids is about 0.93 times the calculated value.

The Term “Osmolarity.” Because of the difficulty of measuring kilograms of water in a solution, which is required to determine osmolality, *osmolarity*, which is the osmolar concentration expressed as *osmoles per liter of solution* rather than osmoles per kilogram of water, is used instead. Although, strictly speaking, it is osmoles per kilogram of water (osmolality) that determines osmotic pressure, for dilute solutions such as those in the body, the quantitative differences between osmolarity and osmolality are less than 1 per cent. Because it is far more practical to measure osmolarity than osmolality, this is the usual practice in almost all physiologic studies.

“Active Transport” of Substances Through Membranes

At times, a large concentration of a substance is required in the intracellular fluid even though the extracellular fluid contains only a small concentration. This is true, for instance, for potassium ions. Conversely, it is important to keep the concentrations of other ions very low inside the cell even though their concentrations in the extracellular fluid are great. This is especially true for sodium ions. Neither of these two effects could occur by simple diffusion, because simple diffusion eventually equilibrates concentrations on the two sides of the membrane. Instead, some energy source must cause excess movement of potassium ions to the inside of cells and excess movement of sodium ions to the outside of cells. When a cell membrane moves molecules or ions “uphill” against a concentration gradient (or “uphill” against an electrical or pressure gradient), the process is called *active transport*.

Different substances that are actively transported through at least some cell membranes include sodium ions, potassium ions, calcium ions, iron ions, hydrogen ions, chloride ions, iodide ions, urate ions, several different sugars, and most of the amino acids.

Primary Active Transport and Secondary Active Transport.

Active transport is divided into two types according to the source of the energy used to cause the transport: *primary active transport* and *secondary active transport*. In primary active transport, the energy is derived directly from breakdown of adenosine triphosphate (ATP) or of some other high-energy phosphate compound. In secondary active transport, the energy is derived secondarily from energy that has been stored in the form of ionic concentration differences of secondary molecular or ionic substances between the two sides of a cell membrane, created originally by primary active transport. In both instances, transport depends on *carrier proteins* that penetrate through the cell membrane, as is true for facilitated diffusion. However, in active transport, the carrier protein functions differently from the carrier in facilitated diffusion because it is capable of imparting energy to the transported substance to move it against the electrochemical gradient. Following are some examples of primary active transport and secondary active transport, with more detailed explanations of their principles of function.

Primary Active Transport**Sodium-Potassium Pump**

Among the substances that are transported by primary active transport are sodium, potassium, calcium, hydrogen, chloride, and a few other ions.

The active transport mechanism that has been studied in greatest detail is the *sodium-potassium* ($\text{Na}^+\text{-K}^+$) pump, a transport process that pumps sodium ions outward through the cell membrane of all cells and at the same time pumps potassium ions from the outside to the inside. This pump is responsible for maintaining the sodium and potassium concentration differences across the cell membrane, as well as for establishing a negative electrical voltage inside the cells. Indeed, Chapter 5 shows that this pump is also the basis of nerve function, transmitting nerve signals throughout the nervous system.

Figure 4-11 shows the basic physical components of the $\text{Na}^+\text{-K}^+$ pump. The *carrier protein* is a complex of two separate globular proteins: a larger one called the α subunit, with a molecular weight of about 100,000, and a smaller one called the β subunit, with a molecular weight of about 55,000. Although the function of the smaller protein is not known (except that it might anchor the protein complex in the lipid membrane), the larger protein has three specific features that are important for the functioning of the pump:

1. It has three *receptor sites for binding sodium ions* on the portion of the protein that protrudes to the inside of the cell.
2. It has two *receptor sites for potassium ions* on the outside.
3. The inside portion of this protein near the sodium binding sites has ATPase activity.

To put the pump into perspective: When two potassium ions bind on the outside of the carrier protein

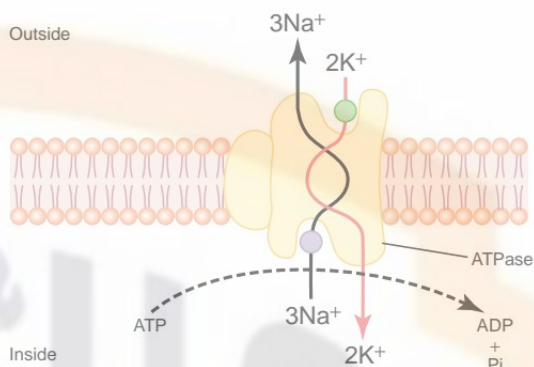


Figure 4-11

Postulated mechanism of the sodium-potassium pump. ADP, adenosine diphosphate; ATP, adenosine triphosphate; P_i , phosphate ion.

and three sodium ions bind on the inside, the ATPase function of the protein becomes activated. This then cleaves one molecule of ATP, splitting it to adenosine diphosphate (ADP) and liberating a high-energy phosphate bond of energy. This liberated energy is then believed to cause a chemical and conformational change in the protein carrier molecule, extruding the three sodium ions to the outside and the two potassium ions to the inside.

As with other enzymes, the $\text{Na}^+\text{-K}^+$ ATPase pump can run in reverse. If the electrochemical gradients for Na^+ and K^+ are experimentally increased enough so that the energy stored in their gradients is greater than the chemical energy of ATP hydrolysis, these ions will move down their concentration gradients and the $\text{Na}^+\text{-K}^+$ pump will synthesize ATP from ADP and phosphate. The phosphorylated form of the $\text{Na}^+\text{-K}^+$ pump, therefore, can either donate its phosphate to ADP to produce ATP or use the energy to change its conformation and pump Na^+ out of the cell and K^+ into the cell. The relative concentrations of ATP, ADP, and phosphate, as well as the electrochemical gradients for Na^+ and K^+ , determine the direction of the enzyme reaction. For some cells, such as electrically active nerve cells, 60 to 70 per cent of the cells' energy requirement may be devoted to pumping Na^+ out of the cell and K^+ into the cell.

Importance of the $\text{Na}^+\text{-K}^+$ Pump for Controlling Cell Volume.

One of the most important functions of the $\text{Na}^+\text{-K}^+$ pump is to control the volume of each cell. Without function of this pump, most cells of the body would swell until they burst. The mechanism for controlling the volume is as follows: Inside the cell are large numbers of proteins and other organic molecules that cannot escape from the cell. Most of these are negatively charged and therefore attract large numbers of potassium, sodium, and other positive ions as well. All these molecules and ions then cause osmosis of water to the interior of the cell. Unless this is checked, the

cell will swell indefinitely until it bursts. The normal mechanism for preventing this is the $\text{Na}^+\text{-K}^+$ pump. Note again that this device pumps three Na^+ ions to the outside of the cell for every two K^+ ions pumped to the interior. Also, the membrane is far less permeable to sodium ions than to potassium ions, so that once the sodium ions are on the outside, they have a strong tendency to stay there. Thus, this represents a net loss of ions out of the cell, which initiates osmosis of water out of the cell as well.

If a cell begins to swell for any reason, this automatically activates the $\text{Na}^+\text{-K}^+$ pump, moving still more ions to the exterior and carrying water with them. Therefore, the $\text{Na}^+\text{-K}^+$ pump performs a continual surveillance role in maintaining normal cell volume.

Electrogenic Nature of the $\text{Na}^+\text{-K}^+$ Pump. The fact that the $\text{Na}^+\text{-K}^+$ pump moves three Na^+ ions to the exterior for every two K^+ ions to the interior means that a net of one positive charge is moved from the interior of the cell to the exterior for each cycle of the pump. This creates positivity outside the cell but leaves a deficit of positive ions inside the cell; that is, it causes negativity on the inside. Therefore, the $\text{Na}^+\text{-K}^+$ pump is said to be *electrogenic* because it creates an electrical potential across the cell membrane. As discussed in Chapter 5, this electrical potential is a basic requirement in nerve and muscle fibers for transmitting nerve and muscle signals.

Primary Active Transport of Calcium Ions

Another important primary active transport mechanism is the *calcium pump*. Calcium ions are normally maintained at extremely low concentration in the intracellular cytosol of virtually all cells in the body, at a concentration about 10,000 times less than that in the extracellular fluid. This is achieved mainly by two primary active transport calcium pumps. One is in the cell membrane and pumps calcium to the outside of the cell. The other pumps calcium ions into one or more of the intracellular vesicular organelles of the cell, such as the sarcoplasmic reticulum of muscle cells and the mitochondria in all cells. In each of these instances, the carrier protein penetrates the membrane and functions as an enzyme ATPase, having the same capability to cleave ATP as the ATPase of the sodium carrier protein. The difference is that this protein has a highly specific binding site for calcium instead of for sodium.

Primary Active Transport of Hydrogen Ions

At two places in the body, primary active transport of hydrogen ions is very important: (1) in the gastric glands of the stomach, and (2) in the late distal tubules and cortical collecting ducts of the kidneys.

In the gastric glands, the deep-lying *parietal cells* have the most potent primary active mechanism for transporting hydrogen ions of any part of the body. This is the basis for secreting hydrochloric acid in the stomach digestive secretions. At the secretory ends of the gastric gland parietal cells, the hydrogen ion concentration is increased as much as a millionfold and then released into the stomach along with chloride ions to form hydrochloric acid.

In the renal tubules are special *intercalated cells* in the late distal tubules and cortical collecting ducts that also transport hydrogen ions by primary active transport. In this case, large amounts of hydrogen ions are secreted from the blood into the urine for the purpose of eliminating excess hydrogen ions from the body fluids. The hydrogen ions can be secreted into the urine against a concentration gradient of about 900-fold.

Energetics of Primary Active Transport

The amount of energy required to transport a substance actively through a membrane is determined by how much the substance is concentrated during transport. Compared with the energy required to concentrate a substance 10-fold, to concentrate it 100-fold requires twice as much energy, and to concentrate it 1000-fold requires three times as much energy. In other words, the energy required is proportional to the *logarithm* of the degree that the substance is concentrated, as expressed by the following formula:

$$\text{Energy (in calories per osmole)} = 1400 \log \frac{C_1}{C_2}$$

Thus, in terms of calories, the amount of energy required to concentrate 1 osmole of substance 10-fold is about 1400 calories; or to concentrate it 100-fold, 2800 calories. One can see that the energy expenditure for concentrating substances in cells or for removing substances from cells against a concentration gradient can be tremendous. Some cells, such as those lining the renal tubules and many glandular cells, expend as much as 90 per cent of their energy for this purpose alone.

Secondary Active Transport—Co-Transport and Counter-Transport

When sodium ions are transported out of cells by primary active transport, a large concentration gradient of sodium ions across the cell membrane usually develops—high concentration outside the cell and very low concentration inside. This gradient represents a storehouse of energy because the excess sodium outside the cell membrane is always attempting to diffuse to the interior. Under appropriate conditions, this diffusion energy of sodium can pull other substances along with the sodium through the cell membrane. This phenomenon is called *co-transport*; it is one form of *secondary active transport*.

For sodium to pull another substance along with it, a coupling mechanism is required. This is achieved by means of still another carrier protein in the cell membrane. The carrier in this instance serves as an attachment point for both the sodium ion and the substance to be co-transported. Once they both are attached, the energy gradient of the sodium ion causes both the sodium ion and the other substance to be transported together to the interior of the cell.

In *counter-transport*, sodium ions again attempt to diffuse to the interior of the cell because of their large concentration gradient. However, this time, the substance to be transported is on the inside of the cell and must be transported to the outside. Therefore, the sodium ion binds to the carrier protein where it projects to the exterior surface of the membrane, while the substance to be counter-transported binds to the interior projection of the carrier protein. Once both have bound, a conformational change occurs, and energy released by the sodium ion moving to the interior causes the other substance to move to the exterior.

Co-Transport of Glucose and Amino Acids Along with Sodium Ions

Glucose and many amino acids are transported into most cells against large concentration gradients; the mechanism of this is entirely by co-transport, as shown in Figure 4-12. Note that the transport carrier protein has two binding sites on its exterior side, one for sodium and one for glucose. Also, the concentration of sodium ions is very high on the outside and very low inside, which provides energy for the transport. A special property of the transport protein is that a conformational change to allow sodium movement to the interior will not occur until a glucose molecule also attaches. When they both become attached, the conformational change takes place automatically, and the sodium and glucose are transported to the inside of the cell at the same time. Hence, this is a *sodium-glucose co-transport* mechanism.

Sodium co-transport of the amino acids occurs in the same manner as for glucose, except that it uses a different set of transport proteins. Five *amino acid transport proteins* have been identified, each of which is responsible for transporting one subset of amino acids with specific molecular characteristics.

Sodium co-transport of glucose and amino acids occurs especially through the epithelial cells of the intestinal tract and the renal tubules of the kidneys to promote absorption of these substances into the blood, as is discussed in later chapters.

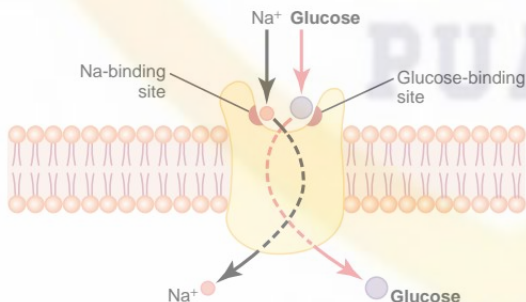


Figure 4-12

Postulated mechanism for sodium co-transport of glucose.

Other important co-transport mechanisms in at least some cells include co-transport of chloride ions, iodine ions, iron ions, and urate ions.

Sodium Counter-Transport of Calcium and Hydrogen Ions

Two especially important counter-transport mechanisms (transport in a direction opposite to the primary ion) are *sodium-calcium counter-transport* and *sodium-hydrogen counter-transport*.

Sodium-calcium counter-transport occurs through all or almost all cell membranes, with sodium ions moving to the interior and calcium ions to the exterior, both bound to the same transport protein in a counter-transport mode. This is in addition to primary active transport of calcium that occurs in some cells.

Sodium-hydrogen counter-transport occurs in several tissues. An especially important example is in the *proximal tubules* of the kidneys, where sodium ions move from the lumen of the tubule to the interior of the tubular cell, while hydrogen ions are counter-transported into the tubule lumen. As a mechanism for concentrating hydrogen ions, counter-transport is not nearly as powerful as the primary active transport of hydrogen ions that occurs in the more distal renal tubules, but it can transport extremely large numbers of hydrogen ions, thus making it a key to hydrogen ion control in the body fluids, as discussed in detail in Chapter 30.

Active Transport Through Cellular Sheets

At many places in the body, substances must be transported all the way through a cellular sheet instead of simply through the cell membrane. Transport of this type occurs through the (1) intestinal epithelium, (2) epithelium of the renal tubules, (3) epithelium of all exocrine glands, (4) epithelium of the gallbladder, and (5) membrane of the choroid plexus of the brain and other membranes.

The basic mechanism for transport of a substance through a cellular sheet is (1) *active transport* through the cell membrane *on one side* of the transporting cells in the sheet, and then (2) either *simple diffusion* or *facilitated diffusion* through the membrane *on the opposite side* of the cell.

Figure 4-13 shows a mechanism for transport of sodium ions through the epithelial sheet of the intestines, gallbladder, and renal tubules. This figure shows that the epithelial cells are connected together tightly at the luminal pole by means of junctions called "kisses." The brush border on the luminal surfaces of the cells is permeable to both sodium ions and water. Therefore, sodium and water diffuse readily from the lumen into the interior of the cell. Then, at the basal and lateral membranes of the cells, sodium ions are actively transported into the extracellular fluid of the surrounding connective tissue and blood vessels. This creates a high sodium ion concentration gradient across these membranes, which in turn causes osmosis

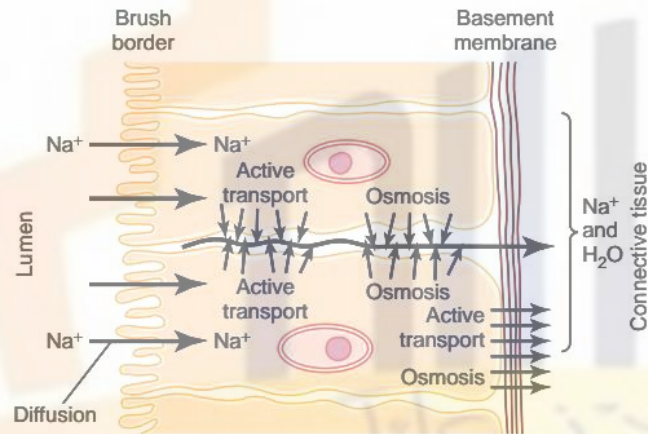


Figure 4-13

Basic mechanism of active transport across a layer of cells.

of water as well. Thus, active transport of sodium ions at the basolateral sides of the epithelial cells results in transport not only of sodium ions but also of water.

These are the mechanisms by which almost all the nutrients, ions, and other substances are absorbed into the blood from the intestine; they are also the way the same substances are reabsorbed from the glomerular filtrate by the renal tubules.

Throughout this text are numerous examples of the different types of transport discussed in this chapter.

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